STEPWISE TEAM

How does a person with Parkinson's walk in daily life, when nobody is watching?

Step monitoring with smart shoes and AI agents for people with Parkinson's disease

60%

of people with Parkinson's fall at least once

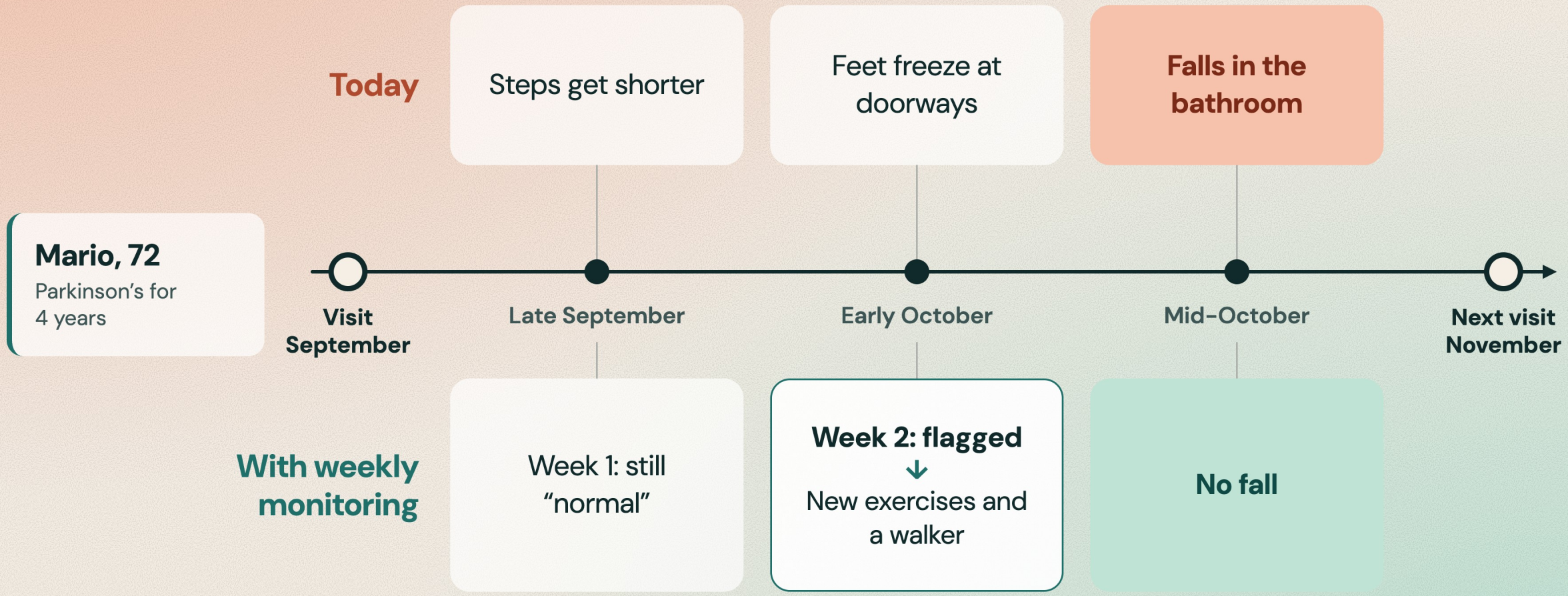
39%

fall repeatedly

Allen et al., 2013, review of 22 studies

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Between two visits, nobody sees how the patient walks



Our idea

HOW
Shoe sensors
step length and duration, variability, freezing

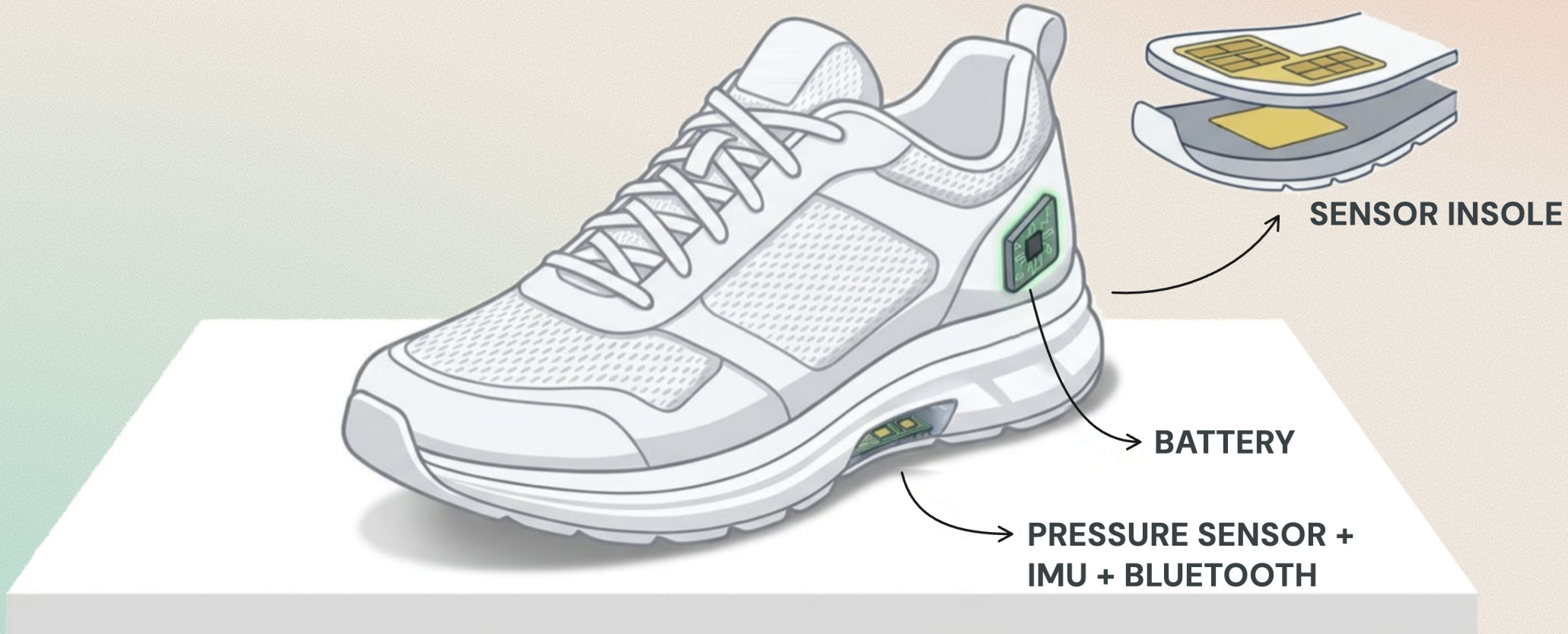
WHAT
Calendar
daily activities and completed exercises

WHERE
GIS location
from the phone: home, uphill, slippery ground

→

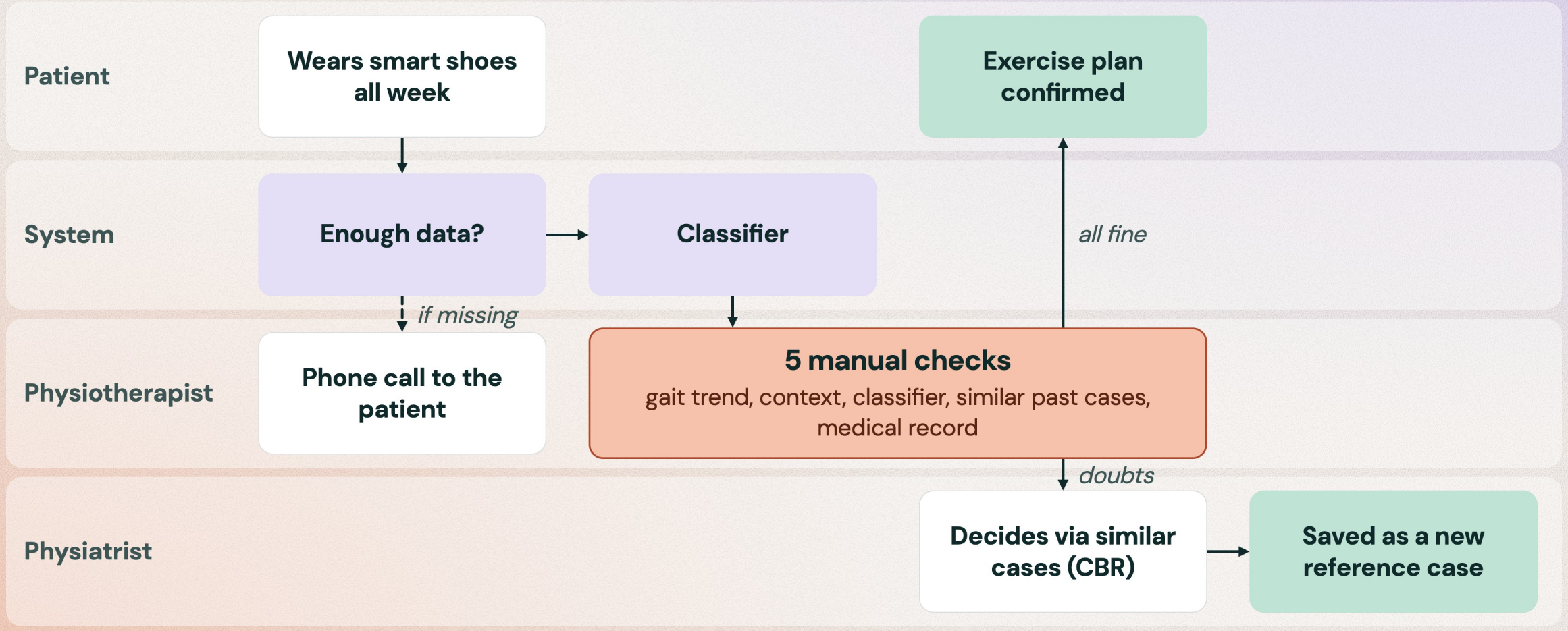
EVERY WEEK
A classifier combines them

normal anomalous



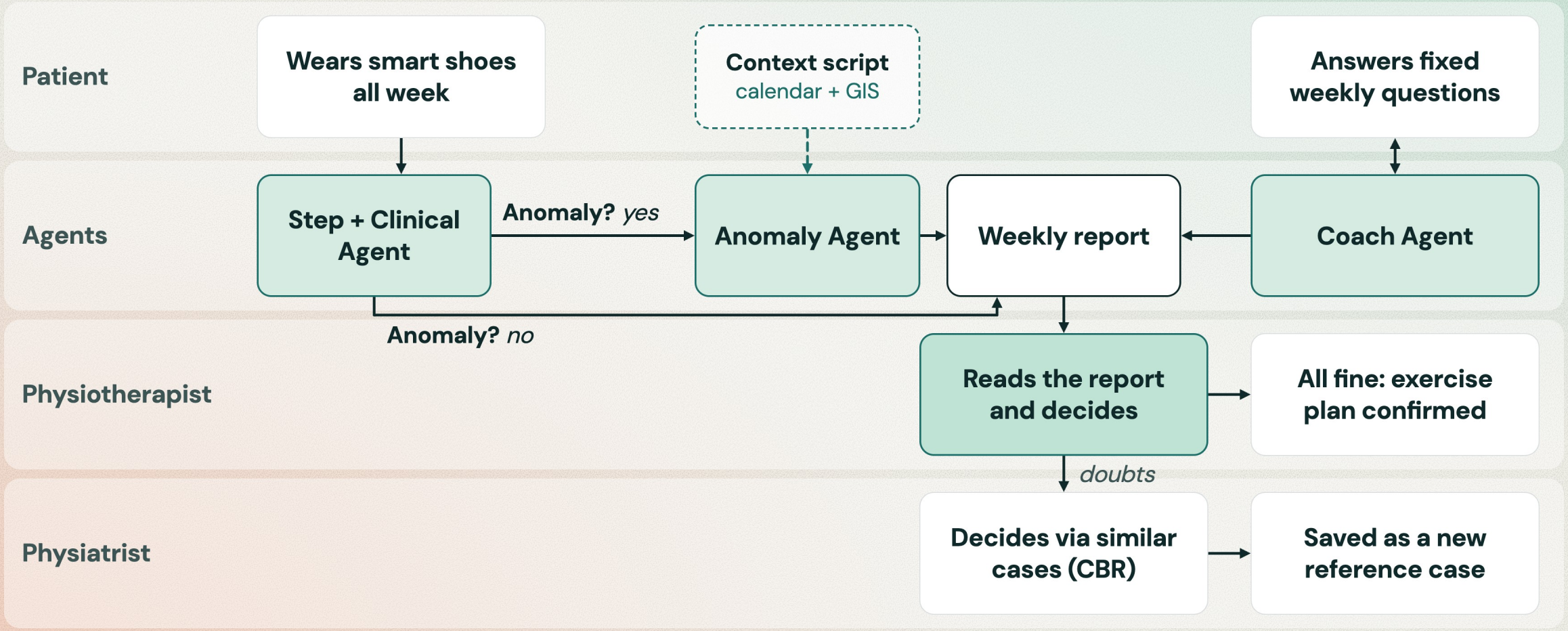
Workflow without agents

The bottleneck
5 manual checks per patient, every week: the workload grows



Workflow with agents

The difference
1 weekly report instead of 5 manual checks



Insights of the agents

Step Agent

Checks the data are reliable (if not, asks for a 2-minute walking test), runs the ML classifier, tracks the trend against the exercise goals.

Tool use | Memory management | Goal setting & monitoring | Exception handling & recovery

Anomaly Agent

Only for anomalous weeks: is the anomaly credible? It uses the context and notes by Clinical Agent. Caution or decline?

Tool use | Reasoning

Clinical Agent

Flags recent changes that may explain the step, such as new drugs or dose timing. It never diagnoses.

Tool use | Reasoning

Coach Agent

Sends only approved messages and asks fixed weekly questions (falls, near-falls, pain).

Prompt chaining | Guardrails | Human-in-the-loop